

This assignment will not be graded and is only for practice.

Level 1

1) Let $f(x, y) = x \log(y)$ and $Hf(x, y)$ denote the Hessian matrix of f at (x, y) . Which of the following options are correct? **1 point**

- ☐ Rank of $Hf(0, 1)$ is 2.
- ☐ Rank of $Hf(0, 1)$ is 1.
- ☐ $(0, 1)$ is a local minimum.
- ☐ $(0, 1)$ is a saddle point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Rank of $Hf(0, 1)$ is 2.
 $(0, 1)$ is a saddle point.

2) Choose the correct statements about $f(x, y) = x^2 + axy + by^2$. **1 point**

- ☐ $(0, 0)$ is a saddle point of f iff $4b - a^2 < 0$.
- ☐ $(0, 0)$ is a saddle point of f iff $4b - a^2 > 0$.
- ☐ There are infinitely many possible pairs (a, b) for which f has local minimum at $(0, 0)$.
- ☐ If $a = b = 1$ then f has a local maximum at $(0, 0)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$ is a saddle point of f iff $4b - a^2 < 0$.
 There are infinitely many possible pairs (a, b) for which f has local minimum at $(0, 0)$.

3) Consider the following function

1 point

$$f(x, y) = x^2 + xy + y^2 + 2x - 2y + 5$$

Which of the following option(s) is (are) true?

- ☐ $(-2, 2)$ is a critical point of f .
- ☐ $(2, -2)$ is a critical point of f .
- ☐ Determinant of the Hessian matrix at point (x, y) is 3.
- ☐ f has a local minimum at point $(-2, 2)$.
- ☐ $(2, -2)$ is a saddle point of f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-2, 2)$ is a critical point of f .

Determinant of the Hessian matrix at point (x, y) is 3.

f has a local minimum at point $(-2, 2)$.

4) For what value of k is the Hessian test inconclusive for the function $f(x, y) = x^3 + kxy + y^3$ at $(0, 0)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2

5) Choose the correct statements about $f(x, y) = 2x^3 - 3x^2y - 12x^2 - 3y^2$.

1 point

- ☐ f has 3 critical points.

- ☐ $(0, 0)$ is a local maximum.
- ☐ $(0, 0)$ is a saddle point.
- ☐ $(-4, -8)$ is a saddle point.
- ☐ $(-4, -8)$ is a local minimum.
- ☐ f has a global minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f has 3 critical points.

$(0, 0)$ is a local maximum.

$(-4, -8)$ is a saddle point.

6) Choose the correct statements about $f(x, y) = x^8 - 2x^4y + y^3 - y$.

1 point

- ☐ The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$.
- ☐ The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & 8x^3 \\ 8x^3 & 6y \end{bmatrix}$.
- ☐ f has four critical points.
- ☐ $(1, 1)$ is a point of local minimum of f .
- ☐ $(1, 1)$ is a point of local maximum of f .
- ☐ Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The Hessian matrix of f at point (x, y) is $\begin{bmatrix} 56x^6 - 24x^2y & -8x^3 \\ -8x^3 & 6y \end{bmatrix}$.

f has four critical points.

$(1, 1)$ is a point of local minimum of f .

Second derivative test is inconclusive for the point $(0, \frac{1}{\sqrt{3}})$.

7) If M denotes the local maximum value of the function $F(a, b) = \int_a^b 3 - 2x - x^2 dx$ over all possible values of a and b then find the value of $3M$. Note that the function has only one local maximum and you have to find that.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 32

1 point

8) Consider a function $f(x, y)$. Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k . Then which of the following options is(are) true? **1 point**

- ☐ If $H_1 = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$, then v_1 is a saddle point.
- ☐ If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.
- ☐ If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.
- ☐ If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

- If $H_2 = \begin{bmatrix} 1 & 1 \\ 3 & 2 \end{bmatrix}$, then v_2 is a saddle point.
- If $H_3 = \begin{bmatrix} 3 & 1 \\ -3 & 1 \end{bmatrix}$, then v_3 is a point of local extremum.
- If $H_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then v_1 is a point of local minimum.

Your score is: 0/8

